

Machine Language and Programming Basics

1. What is machine language (or machine code) composed of, and how does a computer interpret this sequence?
2. Why is it "almost impossible" for humans to program directly using machine language?
3. What is an algorithm, and what is its role for programmers *before* writing instructions in a programming language?

Low-Level and High-Level Languages

4. What is the primary function of Low-Level Languages (LLL) in relation to the computer?
5. What are the two examples of machine-dependent, low-level languages mentioned in the text?
6. What characteristic of programs written in a low-level language allows them to be directly executable on the computer hardware?
7. How does the syntax of High-Level Languages (HLL) differ from low-level languages, and what benefit does this provide to the programmer?
8. What key advantage does a high-level language offer regarding its machine-dependency?

Translation Programs

9. What is the main function of translation programs in the context of programming languages?
10. Describe the two-step process a compiler uses to convert a program into an executable file, mentioning the two types of programs involved.
11. How does an interpreter process the source code differently from a compiler?
12. What is the specific role of an assembler?
13. Which translation program deals specifically with the translation of assembly language?
14. Which translation program creates an object file as an intermediate step?
15. Which programming language, if any, does not require a translation program to be directly executable on the computer hardware?

ALGORITHM

16. Who was Mohammed Ibn-Musa al-Khwarizmi, and what significant contribution did he make that relates to the word "algorithm"?
17. The text defines an algorithm as a set of "precise, i.e. unambiguous, rules." Why is the characteristic of being unambiguous vital for solving a problem?
18. What is the definition of the algorithm characteristic known as correctness?
19. Explain why the analysis of algorithms often resembles "other fields of mathematics" in its approach to study.
20. The text states that algorithms must be defined rigorously. What specifically must this rigorous definition apply to?
21. Which two common types of computer applications mentioned in the text involve modifying the state or security of data (making it smaller or protected)?
22. What specific detail does an algorithm provide in a computer program regarding the steps to be performed?
23. The characteristic of determinism states that intermediate results are determined only by inputs and the results of the previous step. What does this highlight about the process of an algorithm?
24. Name the two distinct ways mentioned in the text that algorithms can be expressed or shown.
25. Which two applications listed in the text are primarily used for organizing and accessing information stored in files or text?

COMPUTER LANGUAGES vs NATURAL LANGUAGES

26. What is the fundamental difference in purpose between natural languages and programming languages, according to the text?
27. Explain the difference between computer languages and natural languages regarding ambiguity (being context-free vs. context-dependent).
28. Why do programming languages require a high degree of precision and expertise from the programmer?
29. What characteristic of computer language commands ensures that the meaning is understood by all computers?
30. In what way are high-level languages similar to natural languages in terms of their structure?
31. How does the ability of a computer to "read between the lines" affect the way minor errors are handled in programming languages compared to natural languages?
32. Is the syntax of programming languages based on the grammar of natural languages?

33. Regarding formality, how do natural languages differ from programming languages?
34. What specific examples are provided to illustrate the creative nature of natural languages?
35. Name three components (rules/elements) that are part of natural languages but are not explicitly listed as components of computer languages (though some may overlap).

WORDPROCESSORS SPREADSHEETS AND DATABASES

36. What is a word processor used for?
37. What are the main functions of a word processor?
38. What is text formatting in a word processor?
39. What is text editing in a word processor?
40. What are some important features of word processors?
41. How has the use of word processors evolved from earlier versions?
42. What are some examples of available word processing software?
43. What is a non-relational database?
44. What is a flat-file database?
45. How is data organized in a network database?
46. What is an object-oriented database?
47. What is a relational database?
48. Why are relational databases predominant today?
49. What is the first step in creating a database?
50. What are the most common data types in databases?
51. What does a database contain?
52. What is metadata in a database?
53. What does metadata include?
54. What is a DBMS (Database Management System)?
55. What are database applications used for?
56. What signs is the goal of data management system?

1.GPS

57. What is a GPS made up of?

- 58. Who owns the GPS system?
- 59. How many satellites are in orbit around the Earth?
- 60. How many satellites are always above the horizon? 4
- 61. What does each satellite contain?
- 62. What is triangulation?
- 63. What are the major uses of GPS technology in science?
- 64. How do we use GPS in our daily lives?
- 65. How does GPS operate?

WEBSITE CREATION

- 66. What are the steps to create a website from scratch?
- 67. What is the first step when building a website from scratch?
- 68. What is a domain name?
- 69. What does WYSIWYG stand for and how does it relate to website creation?
- 70. Why is creating a sitemap important in the website building process?
- 71. What are some of the essential pages of a professional website?
- 72. What is the typical website layout?

INDUSTRIAL AND MANUFACTURING SOFTWARE

- 73. What is an industrial software?
- 74. What is the purpose of manufacturing software?
- 75. What are the different types of manufacturing software?
- 76. What are some activities possible with CAD?
- 77. How does manufacturing software interact with robotics?
- 78. What is robotics in the context of manufacturing?

Industrial Applications and The Fourth Industrial Revolution

79. What is Industry 4.0?
80. Where did the term "Industry 4.0" originate?
81. What is the purpose of Industry 4.0?
82. What is 3D printing, and how does it work?
83. How has 3D printing evolved in Industry 4.0?
84. What is a delivery drone?
85. What are some features of delivery drones?
86. What types of products are typically delivered by drones?
87. Why is drone delivery particularly important in the shipping process?
88. What is the Internet of Things (IoT)?
89. What are the four industrial revolutions?

Communication and telecommunication systems

90. What is communication, and when is it called telecommunications?
91. What is the objective of telecommunications?
92. What are the basic components of a telecommunication system?
93. What is the role of a transducer in telecommunications?
94. What additional components can be implemented in a telecommunication system?
95. How must a message be converted before being sent in a telecommunication system?
96. What is a simplex telecommunication system? Give an example.
97. How does a duplex telecommunication system work?
98. What characterizes a broadcast telecommunication system?
99. What is a multiplex telecommunication system? Provide an example.

METHODS OF TRANSMISSION

100. What are the two main methods of data transmission?
101. What is synchronous transmission, and how does it work?
102. Why is synchronization important in synchronous transmission?

103. What are some examples of synchronous transmission?
104. How does asynchronous transmission differ from synchronous transmission?
105. What are some examples of asynchronous transmission?
106. What is circuit switching in network transmission?
107. What type of communication uses circuit switching, and why?
108. How does packet switching differ from circuit switching?
109. What are the advantages and challenges of packet switching?
110. What is short-region wireless transmission?
111. What is Wi-Fi, and how does it work?
112. What is a Basic Service Set (BSS) in a Wi-Fi network?
113. What role does a Distribution System (DS) play in Wi-Fi networks?
114. What is an Access Point (AP) in a Wi-Fi network?
115. What is Bluetooth technology, and what does it do?
116. How do Bluetooth devices communicate?
117. What is mobile transmission?
118. What is Long-Term Evolution (LTE)?
119. Why do LTE frequencies differ between countries?
120. What is HSDPA, and how does it work?
121. What is the relationship between HSDPA and HSPA+?
122. Why is HSPA+ sometimes mistakenly classified as 4G?
123. What distinguishes 5G NR from previous generations of mobile transmission?
124. What are some advantages of 5G NR compared to earlier mobile transmission standards?

NETWORKS

125. What is a network?
126. What devices can be shared in a network?
127. What are the two types of network operation models?
128. What is a peer-to-peer network?

- 129.What are the advantages of a peer-to-peer network?
- 130.What is a disadvantage of a peer-to-peer network?
- 131.What is a client-server network?
- 132.What are the benefits of a client-server network?
- 133.What is a major disadvantage of a client-server network?
- 134.Which type of network is more suitable for larger setups?

NETWORK COMPONENTS

- 135.What are the media through which data signals can travel in a network?
- 136.What is the purpose of a network interface card (NIC)?
- 137.What is a hub and its function in a network?
- 138.What is the main standard for hardware connectivity in networks?
- 139.What is the role of a router in a network?
- 140.What is a broadband router used for?
- 141.How does a network switch differ from a hub?
- 142.What is the purpose of a bridge in a network?
- 143.What does a repeater do in a network?
- 144.What is the function of a gateway in networking?

TYPES OF NETWORKS

- 145.What does PAN stand for, and what is its range?
- 146.What devices can be connected using a PAN?
- 147.What is a LAN, and where is it typically used?
- 148.What are the common methods of connection in a LAN?
- 149.What is the primary purpose of a MAN?
- 150.What is a WAN, and how does it differ from a LAN?
- 151.What are the primary means of connection in a WAN?
- 152.What is a GAN, and what networks does it include?
- 153.What is a VPN, and what is its primary function?

154.What are the three main types of VPNs and their purposes?

NETWORK TOPOLOGIES

155.What is a network topology?

156.What is the simplest type of network connection?

157.What are the characteristics of a line or line bus topology?

158.What is the main advantage and disadvantage of a bus topology?

159.What is the structure of a star topology?

160.What are the advantages and disadvantages of a star topology?

161.What happens in a ring topology when a node breaks down?

162.What is a tree or star bus topology?

163.What is the main difference between full mesh and partial mesh topologies?

164.What is the role of logical topology in a network?

Communication protocols: the ISO/OSI model

165.What is the definition of a communication protocol?

166.What organization developed the OSI model and why?

167.How many layers are in the OSI model and where does Level 1 begin?

168.Which layer is responsible for converting bits into electrical or optical signals?

169.What is the main function of the Data Link Layer (Layer 2)?

170.Which layer is responsible for routing signals and dividing messages into packets?

171.How does the Session Layer (Layer 5) prevent data loss?

172.Which layer acts as a "translator" and handles encryption?

173.What services are provided by the Application Layer (Layer 7)?

174.What does the Transport Layer (Layer 4) decide regarding data transmission?

TCP/IP

175.What does the acronym TCP/IP stand for?

176.What is the general purpose of the TCP/IP protocol?

177.How does TCP establish communication between two applications?

- 178.What does it mean that IP is a "connectionless" communication protocol?
- 179.How is data handled by the IP protocol for transmission?
- 180.When working together, what are the specific roles of TCP and IP?
- 181.How many layers are in the TCP/IP model?
- 182.What is the direction of data flow when a computer sends information?
- 183.What is the role of Layer 4 (the Application Layer)?
- 184.What are three important protocols found in the TCP/IP Application Layer?

The Internet and its services. The History of the Internet

- 185.What is the definition of the Internet according to the text?
- 186.When is the official "birth date" of the Internet?
- 187.What specific event marked the birth of the Internet in 1983?
- 188.What was the precursor to the Internet created in 1969?
- 189.What type of network was Arpanet?
- 190.Which organizations were involved in the creation of Arpanet?
- 191.When did the most significant rise in the number of Internet users occur?

Connecting to the Internet

- 192.What two things are primarily needed to connect a device to the Internet?
- 193.What does the acronym ISP stand for?
- 194.What are some examples of ISPs mentioned in the text?
- 195.What is the primary function of a modem?
- 196.What does a device receive from the ISP once it is connected?
- 197.. According to the text, what is the Internet described as in relation to websites?

Domain Names and DNS

- 198.What does the acronym URL stand for?
- 199.What does the acronym DNS stand for?

200. How is the DNS described in the text?
201. What is the role of the ISP when you try to reach a website?
202. What happens once a request arrives at the DNS?
203. Why is an IP address necessary during the connection process?
204. Has the number of Internet users decreased since the year 2000?

COMPUTER THREADS

205. What is the primary purpose of malware?
206. Name three ways malware can be classified.
207. How can malicious programs be delivered to a system?
208. What is the main characteristic of adware?
209. Does adware generally have malicious intent?
210. Is spam considered proper malware?
211. What is the definition of spam?
212. What is a "bug" in the context of computer programs?
213. Why are security bugs considered dangerous?
214. What is the significance of a sophisticated malware attack having a link to a hacker-controlled server?

Antivirus Software

215. What is antivirus software primarily used for?
216. What two types of scanning do most antivirus programs include?
217. What does an automatic scan typically check?
218. When would a user utilize the manual scan option?
219. Why do antivirus programs regularly update their database of virus types?
220. Besides viruses, what else do most antivirus programs protect against?
221. Name the three classifications of antivirus software based on their functions.
222. What is the most basic type of antivirus software, and what does it do?
223. What does "malware protection antivirus software" combine?
224. What is typically included in an "antivirus software security suite"?

Firewall

- 225.What is the main purpose of a firewall?
- 226.How does a firewall regulate messages entering or leaving a private network?
- 227.Can firewalls be hardware, software, or both?
- 228.Where are hardware firewalls often found or purchased?
- 229.What is a key characteristic of software firewalls?

Network Security

- 230.What does network security consist of?
- 231.What is the purpose of "identification" in network security?
- 232.What is the purpose of "authentication" in network security?
- 233.What does "authorization" determine in a network security system?

Secure Servers

- 234.Why is it important to use a secure server when transferring personal information over the internet?
- 235.How does a secure server protect data during transfer?
- 236.Name three ways secure websites can be verified.

Copyright

- 237.What is copyright?
- 238.What two types of rights does copyright include?
- 239.What do "economic rights" under copyright refer to?
- 240.What do "moral rights" under copyright include?
- 241.What is the main strategy behind copyleft?
- 242.Who chooses copyleft licenses for creative works?
- 243.What is the benefit of using copyleft licenses for creative works?
- 244.What are cookies?

245.Name two purposes for which cookies are used.

246.What does the EU Cookie Law require when a customer enters a website with cookies?

247.How do some websites often influence user consent regarding cookies?

THE PROFILE OF AN HACKER

248.What is the primary goal of ethical hackers?

249.How do ethical hackers differ from malicious hackers?

250.What is a penetration test, and why do companies conduct them?

251.What kind of vulnerabilities do ethical hackers look for in systems?

252.What skills or knowledge are important for becoming an ethical hacker?

253.What is the significance of obtaining a certification like Certified Ethical Hacker (CEH)?

254.What are the two contrasting types of hackers mentioned in the text?

255.Why is the work of ethical hackers important for society?

256.How does the demand for ethical hackers relate to the rise of cyber crimes?

257.In what ways do ethical hackers contribute to the security of personal data?

Cryptography

258.What is the Caesar Cipher?

259.What type of cipher is the Caesar Cipher?

260.How does the Caesar Cipher work?

261.Who famously used the Caesar Cipher, and for what purpose?

Enigma

262.Who is Alan Turing and where did he join a team of code-breakers?

263.What was the main challenge in deciphering the German Enigma machine's code?

264.How many possible codes could be input into Enigma?

265.How did Alan Turing's approach to breaking the Enigma code differ from the rest of his team?

266. What was a significant challenge Alan faced in developing his machine?

267. Who was Joan Clarke and what role did she play in the team?

268. How did Alan and his team finally manage to crack the Enigma code?

269. What happened to Alan Turing in the final scene of the story?